Cambridge
O Level

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Cambridge Ordinary Level

CHEMISTRY **5070/42**

Paper 4 Alternative to Practical **May/June 2016**

MARK SCHEME

Maximum Mark: 60

Published

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Abbreviations used in the mark scheme

- / separates alternatives within a marking point.
- **or** gives the alternative marking point.
- **Allow / accept** indicates an answer that is less than ideal but which should be marked correct.
- **Ignore** means mark as if the response was not there.
- **Reject** means the response is not given credit
- Ecf means credit a correct statement / working that follows from a previous wrong response.
- Use of brackets in the Answer column indicates that the word(s) is / are ideal but not required to obtain the mark.

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Question	Answer	Marks
1(a)	M1:**Thermometer / bulb** Thermometer / bulb is too low / should be higher / should not touch the beads / should be at entrance to condenser (1) M2: **Receiver / conical flask / C** There should be no bung or cork on **C** / **C** should be open (1)	2
1(b)(i)	Fractionating column	1
1(b)(ii)	Separate components / separate mixture / separate heptane and hexane / separate liquids / stop heptane reaching condenser	1
1(b)(iii)	Condenser	1
1(b)(iv)	(To convert) vapour / gas to liquid **or** liquefy vapour / gas **or** condense vapour / gas	1
1(c)(i)	69 °C	1
1(c)(ii)	Hexane	1
1(d)	M1 Electric (heater) / water bath / hot plate / (1) M2 (components of mixture are) flammable / inflammable(1)	2

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Question	Answer	Marks
2(a)	M1 (aqueous) NaOH / sodium hydroxide (solution) (1) M2 A*l* / aluminium(foil) / Devardas alloy (1) M3 Heat / warm (1) M4 Ammonia **or** **gas** turns litmus blue (1)	4
2(b)	M1 Heat (1) M2 To crystallisation point / saturation (point) (1) M3 Wash **and** dry (crystals) (1)	3
2(c)(i)	21 24 (1) (–)3 (1)	2
2(c)(ii)	Endothermic	1

Question	Answer	Marks
3(a)(i)	1.3(0) g	1
3(a)(ii)	1.62 g	1
3(a)(iii)	0.32 g	1
3(a)(iv)	M1 1.30 / 65 and 0.32 / 16 **or** 0.02 and 0.02 **or** both 1 / 50 (1) M2 ZnO (1)	2
3(b)	Hydrogen (1) Pops in a flame / lighted splint pops / burning splint pops (1)	2

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Question	Answer	Marks
4	**B**	1

Question	Answer	Marks
5	**A**	1

Question	Answer	Marks
6	**C**	1

Question	Answer	Marks
7(a)	Blue to colourless	1
7(b)	28.1 30.9 47.1 0.0 3.5 19.5 28.1 27.4 27.6 (3) Mean titre 27.5 cm^3 (1)	4
7(c)	$0.0025 / 2.5 \times 10^{-3}$	1
7(d)	$0.0025 / 2.5 \times 10^{-3}$	1
7(e)	$0.0909 / 9.09 \times 10^{-2}$	1
7(f)	88	1
7(g)	M1 (M_r of COOH) = 45 **or** 12 + 16 + 16 + 1 **or** 12 + 32 + 1 (1) M2 **n** = 3 (1) M3 $C_4H_8O_2$ (1) M4 butanoic acid / methyl propanoic acid (1)	4

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Question	Answer	Marks
8(a)	Colourless (solution)	1
8(b)	**(Both)** Al^{3+} or Zn^{2+} (ions present)	1
8(c)	Al^{3+} (ions confirmed)	1
8(d)	M1 (dilute) HNO_3 / nitric acid ignore acidify(1) M2 (aq) aqueous solution of $AgNO_3$ / silver nitrate (1) M3 white precipitate(1)	3
8(e)	$AlCl_3$	1

Question	Answer	Marks
9(a)	White	1
9(b)	1.3(0), 1.95, 2.6(0), 2.8(0), 2.8(0)	1
9(c)(i)	All points correct (1) (Only) two intersecting straight lines, one mark for each line (2)	3
9(d)(i)	Value as read from graph (correct to within 0.1) e.g. 3.7	1
9(d)(ii)	Value as read from graph (correct to within 0.025) e.g. 2.8	1
9(d)(iii)	Value as read from graph (correct to within 0.1) e.g. 8.6	1
9(e)	M1 $\frac{10 \times 1.2}{8.6}$ (1) **OR** (moles $BaCl_2$) = $\frac{10 \times 1.2}{1000}$ or = 0.012 (1) M2 1.395 / 1.4 (1) (mol / dm^3)	2

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